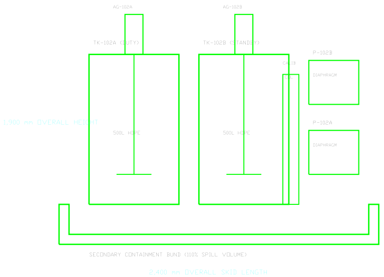

## CHEMICAL REAGENT DOSING SKID SPECIFICATION & BILL OF MATERIALS (SK-100):

ITEM 01: Solution Preparation Tanks TK-102A/B (2 Units, Duty / Standby)

- Net Volume: 500 Liters each | Material: UV-Stabilized Food-Grade Cross-Linked HDPE.
- Tank Dimensions: 800 mm Dia x 1,100 mm Straight Height | Conical bottom with 1.5" drain.
- Accessories: Hinged loading hatch, dust screen, level gauge glass with ultrasonic transmitter.

ITEM 02: High-Shear Agitator Mixers AG-102A/B (2 Units)

- Motor: 0.75 kW, 415V, 3-Phase, IP55, TEFC | Shaft & Impeller: SS316L 4-Blade Marine Propeller.
- Duty: Rapid dissolution of food-grade Sodium Bicarbonate / Citrate salts @ 25-35 deg C.

ITEM 03: Precision Metering Chemical Pumps P-102A/B (2 Units, Duty / Standby)

- Type: Positive Displacement Mechanically Actuated PTFE Diaphragm Pumps.
- Flow Range: 5.0 to 50.0 L/h (Design Operating Point: 9.38 L/h for 0.50 wt% liquor recipe).
- Turndown: 10:1 Turndown ratio with stroke length micrometer + 4-20mA VFD speed modulation.
- Wetted Materials: PVDF Liquid End, Ceramic Double Ball Check Valves, PTFE Diaphragm.

ITEM 04: Skid Piping, Instrumentation & Safety System

- Process Piping: 1" NB Schedule 10S SS316L piping with sanitary tri-clamp connections.
- Safety Relief Valves PSV-102A/B: Set @ 4.0 bar g with full recirculation return to suction tank.
- Pulsation Dampener: Bladder-type 0.5L dampener smoothing discharge flow pulsation to < +/-3%.
- Calibration Column: 1,000 mL Borosilicate glass cylinder with 3-way isolation valve for flow QA.
- Containment: 110% Bund capacity (600L) with optical liquid leakage switch LS-102 interlocked to ESD.

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT

HARAKUTIR AGRO-PROCESS ENGINEERING PVT. LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GENERAL ARRANGEMENT - CHEMICAL DOSING SKID SK-100
DISCIPLINE: MECHANICAL | STAGE: CONCEPTUAL FEED

| | | |
|---|---|---|
| DWG NO: DWG-FIN-15T-009 | REV: V4.0 | SHEET: 9 OF 12 |
| SCALE: AS SHOWN DATE: 2026-10-15 | STATUS: PRELIMINARY | GATE: G-A HOLD |